

Solve the following exercises and problems:

1. Find an optimal parenthesization of a matrix-chain product whose sequence of dimensions is $\langle 20, 6, 50, 10, 3, 20, 25 \rangle$. Show triangular matrices $m[i,j]$ and $s[i,j]$. Show the details of computation for $m[1,6]$, and how to use $s[1,6]$ to obtain the final result.
2. Exercise 15.3-1 page 389. Be brief, but justify your answer.
3. Exercise 15.3-3 page 389. Be brief, but justify your answer.
4. Given

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)!n!} \quad \text{for } n \geq 0.$$

show that the following hold for Catalan numbers:

$$C_0 = 1 \quad \text{and} \quad C_{n+1} = \frac{2(2n+1)}{n+2} C_n,$$

and

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} \quad \text{for } n \geq 1.$$

5. Trace the behaviour of the recursive Hirschberg's quadratic-time linear-space Algorithm C on the strings BBCABA and CBBBAAB. It is enough to show in detail only the main level of recursion of Algorithm C.